

Reproducing ExDBSCAN Counterfactual Clustering Explanations

COMP5331 project type: Implementation

Group number: [Group number]

Members (name; student ID; research/FYP supervisor and topic, if applicable): [Member 1]; [Member 2]; [Member 3]; [Member 4]; [Member 5]; [Member 6, if applicable]

Relationship to members' research/FYP: [For each member, state whether a research/FYP topic exists and, if so, explain how this course project differs.]

Course-project declaration: This work is undertaken for COMP5331 and will be reported as a course project.

Project description

Motivation and objective

Density-based clustering is useful when clusters have irregular shapes and observations may be labeled as noise. DBSCAN makes these decisions through local density and reachability, but an assigned cluster label alone does not explain what would have to change for a particular point to receive a different assignment. Counterfactual explanations turn this into a concrete question: find a small feasible change to an observation that produces a specified alternative outcome. For DBSCAN, this is unusually subtle because moving a point can change not just its nearest cluster but density connectivity, core-point status, and the status of neighboring observations. Matthews and colleagues propose ExDBSCAN to reason about this setting. Our project will reproduce its central explanation-quality experiment on a manageable selection of data and analyze cases where validity, proximity, and diversity pull in different directions.

The main objective is to assess whether the released implementation produces valid and useful counterfactuals under a repeatable DBSCAN setup. We will first reproduce two-dimensional examples, where the original point, candidate counterfactual, cluster borders, and noise region can be inspected visually. We will then evaluate the method on two small public tabular datasets with numerical features. Rather than attempting the entire paper's benchmark collection, we will use a protocol with a fixed query set and explicit parameters, so that each reported explanation can be audited. The outcome will include successful explanations, unsuccessful attempts, and runtime, not only the best examples.

Paper and algorithm scope

The primary paper is Pernille Matthews and coauthors, “ExDBSCAN: Explaining DBSCAN with Counterfactual Reasoning,” KDD 2026; its arXiv version contains additional material. We will use the public author repository as the starting implementation, map its main script and helper modules to the procedure in the paper, and pin the repository revision used for our experiments. We will distinguish the original method from any small compatibility fixes required by current dependencies. The core experiment concerns a single query point and numerical feature changes within the ranges supported by the code. Handling many simultaneous point modifications or domain-specific feasibility constraints is outside the initial scope. This focuses the reproduction on the paper's principal counterfactual objective.

We will define a counterfactual's validity using the same evaluation convention as the paper: after the proposed change, the clustering outcome must meet the designated target or status condition. Before running comparative experiments, we will verify this definition against small synthetic examples. This step is essential because a candidate may look close to another cluster geometrically without satisfying DBSCAN's density-reachability condition. We will also record whether evaluation requires rerunning DBSCAN on the modified data or applying an assignment rule to an otherwise fixed clustering; these are not interchangeable. If the paper and implementation support several target conditions, the report will specify which condition each result addresses.

Data and evaluation design

Our synthetic data will include separated blobs, a curved shape with noise, and a mixture of high- and low-density groups. These examples expose different counterfactual situations: a point near a boundary, a point deep inside a cluster, and a point initially labeled as noise. For the public-data stage, we will use a small UCI or OpenML dataset such as Iris or Wine after choosing numerical columns, applying training-set-derived scaling, and documenting any excluded variables. We will select DBSCAN `eps` and `min_samples` through a transparent preliminary grid based on obtaining a nontrivial clustering. Once parameters are fixed, all compared methods will receive the same clustering, query points, target conditions, and search budget.

We will measure validity rate, counterfactual distance from the original observation, diversity among multiple explanations for one query, and elapsed time per query. Distances will be computed in scaled feature space so that a high-range variable does not dominate by unit choice. We will compare ExDBSCAN with at least a random candidate search and a simple search directed toward points in the desired cluster or noise region. The simple baselines are intentionally limited but reveal whether the paper's reasoning improves over uninformed or geometric alternatives under the same candidate budget. We will report distances only for valid counterfactuals while separately reporting failure rates, preventing a method that fails frequently from appearing unusually close. For each dataset, we will use a fixed set of query points spanning clusters and noise, repeat stochastic components with several seeds, and retain all configuration files.

The principal plots will show example explanations on synthetic data and metric distributions across queries on public data. We will inspect difficult cases individually: points for which no valid alternative is found, explanations that cross a substantial low-density gap, or multiple returned counterfactuals that are nearly identical despite a diversity objective. A sensitivity analysis will change `eps` and `min_samples` within a small neighborhood of the chosen values, since DBSCAN results are parameter-dependent. This is not intended to search for the most favorable score; it tests whether the method's relative behavior is stable when the density definition changes moderately. We will state any dataset or parameter choices that make a target cluster unreachable under the allowed perturbation constraints.

Deliverables and expected insight

We will validate the metric implementation independently on hand-designed point configurations. A known valid candidate, a candidate that lies close to a cluster but remains density-unreachable, and a candidate that changes a neighboring core status will form unit-scale examples for the evaluator. This prevents explanation validity from being inferred solely from a plotted position. We will also save the original and modified cluster labels for each query, permitting later review of whether a method changed only the intended point or altered surrounding cluster structure. The resulting diagnostic cases will guide interpretation of aggregate scores.

The project will deliver a runnable experiment script, pinned environment information, prepared data and query configurations, evaluation code, and regenerated figures and tables. The report will explain the algorithmic definition of a valid DBSCAN counterfactual and compare the released implementation with its stated objective. A successful reproduction should show where ExDBSCAN improves validity or proximity and where its additional reasoning costs runtime. If some public-data cases do not reproduce the paper's advantages, we will investigate cluster geometry, noise proportion, feature scaling, and target feasibility as possible explanations. The scope is deliberately small enough for CPU experimentation while preserving the distinctive scientific question of counterfactual explanation for density-based clustering.

Papers to read

1. Pernille Matthews et al. "ExDBSCAN: Explaining DBSCAN with Counterfactual Reasoning." KDD 2026. [Extended paper](#) · [Author implementation](#).